Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/19/2024
PI Name:	Simpson, Richard Ph.D.		
Project Title:	Exercise Training as Countermeasure for Anti-Tumor Immune Dysregulation Due to Isolation and Confinement Stress		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rjsimpson@email.arizona.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-397-0121
Organization Name:	University of Arizona		
PI Address 1:	College of Agriculture and Life Sciences; College of Medicine		
PI Address 2:	1177 E. Fourth Street, Room 308, Shantz Building		
PI Web Page:			
City:	Tucson	**State:**	AZ
Zip Code:	85721-0001	**Congressional District:**	3
Comments:	NOTE: Formerly at University of Houston until September 2017 move to University of Arizona.		
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	02/01/2021	**End Date:**	12/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per L. Juliette/JSC and NSSC information (Ed., 8/7/23). NOTE: End date changed to 04/06/2023 per NSSC information (Ed., 6/7/22). NOTE: End date changed to 05/15/2023 per L. Juliette/JSC (Ed., 5/8/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Katsanis, Emmanuel M.D. (University of Arizona)		
Grant/Contract No.:	80NSSC21K0452		
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Performance Goal Text:	
Task Description:	Immune dysregulation due to spaceflight may leave future space explorers at a greater risk of latent viral reactivation and hematological malignancies (due to combined radiation exposure). We recently showed that a ~6-month mission to the International Space Station (ISS) impaired the anti-leukemic activity of natural killer cells (NK-cells) by ~50% in vitro compared to pre-flight values and ground-based controls. This response was more marked in first time 'rookie' compared to 'veteran' crew, indicating that it might be a stress-related phenomenon. We further showed that higher pre-flight fitness levels and post flight conditioning (namely VO2max and upper body muscular endurance) was associated with a 29-39% reduction in the incidence of latent viral reactivation, indicating that exercise training might help mitigate some aspects of the spaceflight environment on immune dysregulation. Central Objectives: The overarching objective of this ground-based project is to determine the effects of exercise training as a countermeasure for anti-tumor immune dysregulation due to isolation and confinement stress. We will achieve this objective with the following specific aims: (1) Determine the effects of an isolated and confined spaceflight analog environment (Human Exploration Research Analog : HERA) on the anti-leukemic function of human NK-cells in vivo using a leukemia-bearing humanized NOD-scid-IL2R?null-huIL-15 (NSG-15) mouse model; and (2) Investigate the effects of combined aerobic and resistance exercise training as a potential countermeasure to mitigate the anticipated negative effects of a HERA mission on NK-cell function in vivo. We hypothesize that NK-cells obtained from HERA participants during the mission will have a reduced ability to control leukemic growth in the NSG-15 mice compared to NK-cells collected from the same participants before the mission. We further posit that HERA crewmembers exposed to exercise training will demonstrate smaller declines in NK-cell function compared to crewmembers who are not prescribed exercise. Approach: We will use our established humanized NOD-scid-IL2R?null-huIL-15 (NSG-15) mouse model that allows us to track the growth of human leukemias by bioluminescence imaging with and without human peripheral blood mononuclear cells (PBMCs) transfer. We will also investigate if exercise training can protect HERA crewmembers from impaired NK-cell function by comparing exercised to control HERA participants using the same humanized mouse model as an in vivo endpoint. This project will serve as a daughter-study of our currently-funded NASA project (STUDY#0194) scheduled for HERA campaigns 6 and 7 (C6+C7). Crewmembers in HERA C6 will be prescribed combined aerobic and resistance exercise training in accordance with the parent project, while C7 crewmembers will serve as the control group. To achieve the aims of the present daughter-study, we will request an additional blood aliquot from our current sampling schedule (pre-mission, day 10, day 25, day 40, and post-mission time points) for PBMC cryopreservation and subsequent transfer to luciferase-tagged K562 leukemia bearing NSG-15 mice. Significance of the Proposed work: The current study will address Integrated Research Plan (IRP) gap: "Risk Of Adverse Health Event Due To Altered Immune Response," allowing us to determine if immune dysregulation seen during flight is associated with isolation/confinement stress provided by the HERA analog. We will also determine if changes in NK-cell function that have been documented in vitro translate in vivo using our novel NSG-15 humanized mouse model. Finally, this project will allow us to determine if exercise training can mitigate the effects of isolation/confinement stress to maintain NK-cell function and better control leukemic growth in vivo. This would provide strong support for exercise as an anti-stress immune system countermeasure during future exploration class missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will allow us to determine if exercise training can mitigate the deleterious effects of isolation/confinement stress on anti-tumor immunity. Findings will be of relevance to cancer patients on Earth as exercise is currently recommended as a method to reduce cancer risk and slow disease progression.
Task Progress:	NASA considers spaceflight an isolated, confined, and extreme environment (ICE), which consequently embodies multiple hazards that may impair both the physical and psychological well-being of the astronauts, implying that robust countermeasures to minimize health risks to the crew are vital. Immunity may be dysregulated due to various ICE factors, particularly chronic stress. This is a critical concern as it amplifies health risks such as viral reactivation and impaired anti-tumor immune surveillance, especially in combination with other ICE aspects. An abundance of literature has shown that exercise has a mitigating effect on chronic stress and can also preserve and improve the normal functioning of the immune system, but it is not known if exercise performed in an ICE analog can impact potential changes in biomarkers of stress and immune function. The aim of this study was to determine the effects of a 45-day terrestrial Human Exploration Research Analog (HERA) mission on immune cell phenotypes and function (NK-cell cytotoxicity), blood biomarkers of stress (cortisol, epinephrine, norepinephrine, DHEA-S) and cognitive function (BDNF). We also sought to determine voluntary exercise patterns among the crew and their associations with stress and immune biomarkers. Surprisingly, we found that biomarkers of stress and cognitive function did not change significantly in response to the HERA mission. There was, however, a decrease in NK-cell cytotoxicity just before the late stage of the mission, which is similar to a previous documentation of astronauts aboard the International Space Station. Using a xenogeneic mouse model, we found that peripheral blood mononuclear cells (PBMCs) collected after the mission were more effective at controlling K562 leukemic growth in vivo, which may indicate improvements in immunity following the withdrawal of a stressor. The crew were found to perform aerobic exercise at a consistent volume (intensity x duration) throughout the mission, which resulted in discernible improvements in cardiorespiratory fitness (VO2max). No correlations were found between the change in aerobic fitness or the cumulative exercise volume and the observed decline in NK-cell cytotoxicity. We conclude that a 45-day HERA mission may not accurately capture the stressful environment experienced by astronauts during spaceflight but can cause transient immune dysregulation. Additional data acquired over future HERA missions are required to fully understand the impact that prescribed exercise training may have on immune function during ICE.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Batatinha H, Diak DM, Niemiro GM, Baker FL, Smith KA, Zúñiga TM, Mylabathula PL, Seckeler MD, Lau B, LaVoy EC, Gustafson MP, Katsanis E, Simpson RJ. "Human lymphocytes mobilized with exercise have an anti-tumor transcriptomic profile and exert enhanced graft-versus-leukemia effects in xenogeneic mice." Front Immunol. 2023 Apr 03;14:1067369. https://doi.org/10.3389/fimmu.2023.1067369 ; PMID: 37077913; PMCID: PMC10109447 , Apr-2023